

ONLINE COMPLAINT REGISTRATION SYSTEM

Project Documentation

Submitted By

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Full Stack Development (MERN)

Internship Project

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1. Introduction

- **Project Title:** Online Registration Complaint System
- **Team Members:** KACHANA JAYA KEERTHANA (Individual Submission)-Full Stack Developer
- The Online Complaint Registration System is a web-based application developed using the MERN Stack. It allows users to register complaints online through a simple and user-friendly interface. The system helps reduce manual paperwork by maintaining complaint records digitally. Users can securely log in, submit complaints, and manage complaint details efficiently.

2. Project Overview

- **Purpose:** The goal of this project is to provide a platform for users to easily register their complaints online and for administrators to manage those complaints efficiently.
- **Features:** Key functionalities include user login/registration, complaint submission (Create), tracking complaint status (Read/Update), and an administrative dashboard for oversight.

3. Architecture

- **Frontend:** The frontend is built using React.js for a responsive user interface.
- **Backend:** The backend utilizes Node.js and Express.js to manage API endpoints and server logic.
- **Database:** MongoDB is used for data storage and management.

4. Setup Instructions

- **Prerequisites:** Requires Node.js (v18+), MongoDB (Compass/Atlas), and VS Code.
- **Installation:**
 - Clone the repository using `git clone <your-repo-url>`.
 - Install dependencies by running `npm install`.
 - Set up your MongoDB connection string within the `.env` file.

5. Folder Structure

- **Client:** Contains the src folder, which holds components, pages (e.g., Login, Dashboard), and API service calls.
- **Server:** Organized into controllers, models (for database schema), routes (for API paths), and the server.js entry file.

6. Running the Application

- **Frontend:** Navigate to the client directory and run npm start.
- **Backend:** Navigate to the server directory and run npm start.

7. API Documentation

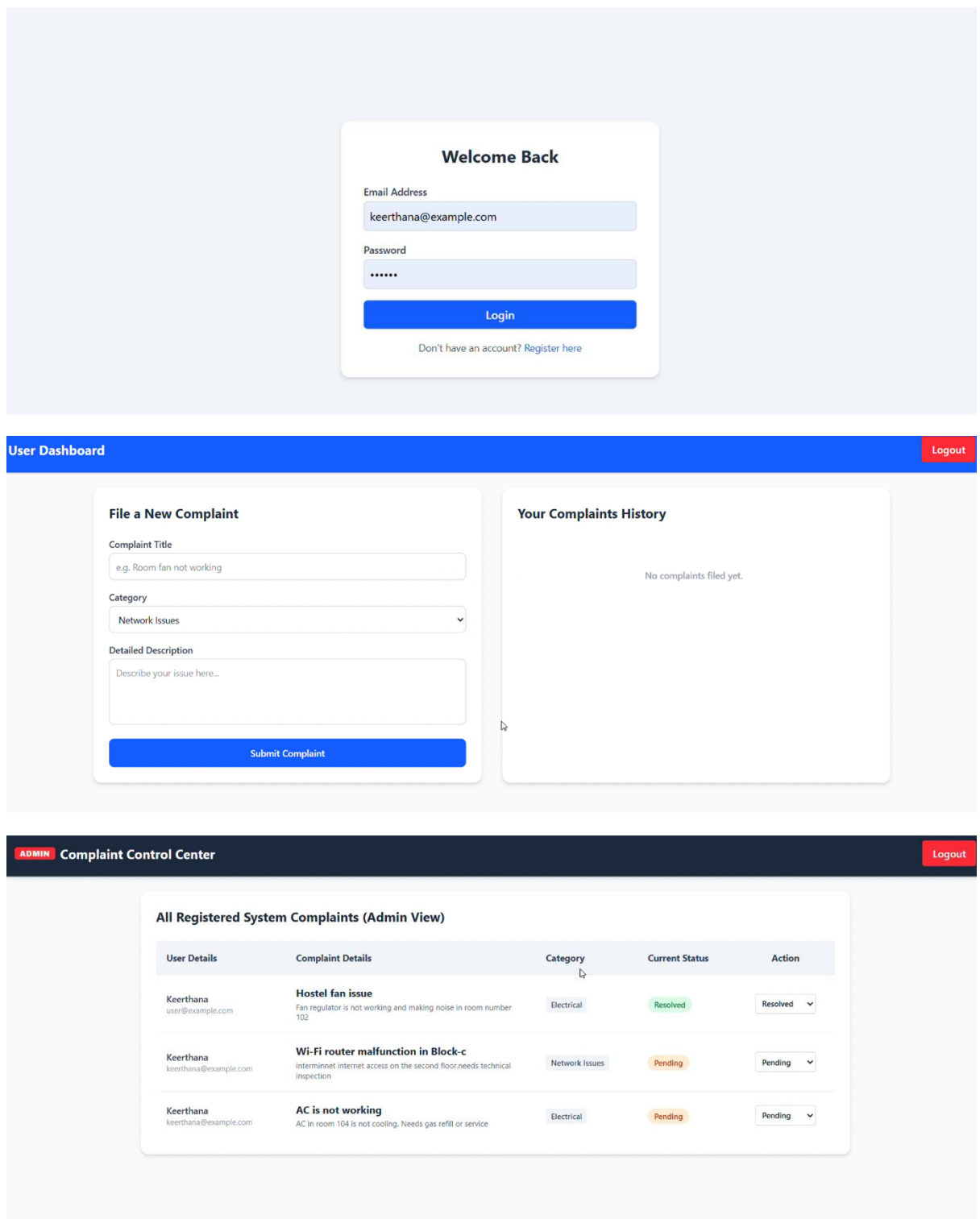
- POST /api/register – Endpoint for new user registration.
- POST /api/login – Endpoint for user authentication.
- POST /api/complaints – Endpoint for submitting a new complaint.
- GET /api/complaints – Endpoint to retrieve a list of all complaints

8. Authentication

- The project uses JSON Web Tokens (JWT) to manage user sessions and ensure secure authorization. Each protected request is verified through the authorization header.

9. User Interface

- **UI Features:** The interface features a clean, responsive design using CSS, ensuring accessibility on both desktop and mobile devices.



10. Testing

- **Strategy:** Manual testing was performed for the complete user journey, including registration, complaint submission, and status updates.

- **Tools:** Postman was utilized to test the functionality and response of the backend API endpoints.

11. Screenshots or Demo

- **Demo:** The live source code and project implementation can be accessed on GitHub at: <https://github.com/Keerthana11309/online-complaint-registration-system>
- **Visuals:** You can find screenshots of the application, including the login flow, complaint submission form, and administrative dashboard, within the repository's documentation folder

12. Known Issues

- **Real-time Updates:** The application does not currently feature a real-time notification system, requiring users to manually refresh their dashboards to view status updates.
- **Session Management:** Developers and users should note that the system requires users to re-login if the current JSON Web Token (JWT) expires.

13. Future Enhancements

- **Real-time Notifications:** Future iterations could integrate Socket.io to provide users with instant alerts regarding their complaint status.
- **User Management:** Planned improvements include adding comprehensive user profile management features.
- **Communication Features:** Future updates may include automated email alerts to notify users whenever their complaint status changes.
- **Admin Dashboard:** The administrative dashboard could be enhanced with advanced search and filtering mechanisms to improve efficiency.

